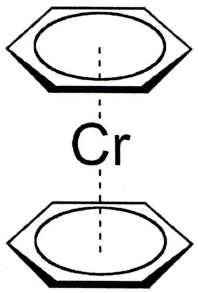
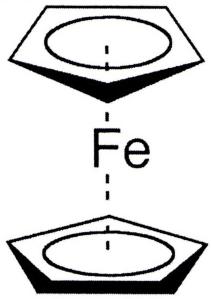
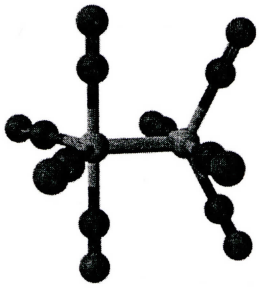
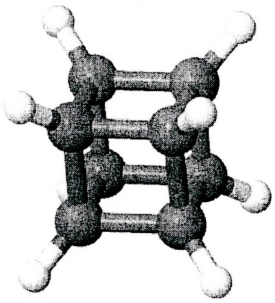
Inorganic and Materials Cumulative Exam
9/7/2018

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. Identify the point group for each of the following molecules (4pts each)

(a) Dibenzene chromium 	(b) Ferrocene 	(c) Dicobalt octacarbonyl 
(d) Cubane (C_8H_8) 	(e) Ammonia (NH_3)	

2. (a) (10pts) Draw the Lewis structures, including any lone pairs, of AsH_3, AsF_3, and AsF_5.
(b) (10pts) Identify each as a Lewis acid or base and explain your answers.

3. (a) (10pts) How are charge-transfer (electronic) transitions (metal to ligand OR ligand to metal) different than the d-d transitions for metal-ligand complexes?
(b) (10pts) Why are charge-transfer transitions often significantly more intense in UV-Vis absorption spectra than the d-d transitions?

4. For hydrogen, the 1s orbital energy is -13.6eV. For fluorine, the 2s orbital energy is -40.2eV and the 2p orbital energy is -18.7eV.
(a) (10pts) Sketch the molecular orbital diagram for HF and identify the bond order.
(b) (6pts) Which of the F valence orbitals are non-bonding? Why?
(c) (4pts) Based on your MO diagram, is the HF^- anion more or less stable than the HF^+ cation? Why?

5. (20pts) Draw all isomers of the anion diaquabisoxalato manganate(III) (charge = -1). Label the isomeric geometry of each. (If you are unsure about the chemical structure of a ligand, you can draw a more generic molecule – one with correct denticity for example - for a small penalty).

6. (a) (4pts each) Sketch the general shape for the temperature dependence of the magnetic susceptibility, χ, for a ferromagnetic, a paramagnetic, and an antiferromagnetic material.

(b) (4pts each) Label the Curie point and the Néel point on your sketches.

7. (4pts each) Assume you are given a crystal of unknown composition. Describe what techniques you would use to determine the following properties of the material experimentally. Please provide a brief description of how the measurement would determine the properties.

(a) Lattice constant and symmetry
(b) Composition
(c) Electrical properties (insulator vs. conductor vs. semiconductor)
(d) Band gap
(e) Band structure

8. (a) (10pts) Describe the construction and operating principles of a semiconductor light emitting diode (LED).

(b) (4pts) What determines the color of an LED?

(c) (6pts) For the design of an LED, is it better for the semiconductor to have a direct or indirect band gap (2pt)? Why (4pt)?

9. (a) (10pts) CsCl forms a cubic lattice with a lattice constant of a=0.4123nm. The unit cell has a Cl^- ion at [0,0,0] and a Cs+ ion at [a/2, a/2, a/2]. What is the density of CsCl?

(b) (10pts) CsCl and NaCl have identical stoichiometry and are both ionic crystals. At ambient temperature and pressure, however, they have different crystal structures (NaCl forms the 'rock salt' crystal symmetry rather than the cubic symmetry). Why do they form different crystal lattices?

10. (a) (5 pts) How is a gel different from a solution?

(b) (5 pts) Sol or gel: which has a higher shear modulus? Please provide a brief explanation of your answer.

(c) (10 pts) What is the difference between a *chemical* gel and a *physical* gel? Please provide on example of each.